

SC 651 - Kalman Filtering - Roll Angle Estimate

Name:

Roll no.:

1 Hardware setup

The given hardware setup contains a breadboard mounted with an ESP32-WROVER B microcontroller and an LSM9DS1 9-DOF sensor module. Sensor and microcontroller connections are already made with wires. The LSM9DS1 sensor module contains a 3-axis gyroscope, a 3-axis accelerometer, and a 3-axis magnetometer. I2C protocol is used for sensor-microcontroller communication (the sensor module also supports the SPI protocol). We will be using Arduino IDE for programming ESP32 (you can use any suitable IDE such as VS Code with platformIO).

2 Kalman Filtering for Roll Angle Estimate

2.1 System Model

Our state is the roll angle (1-dimensional). The system dynamics can be written as:

$$\dot{\theta} = \omega \quad \text{System model} \quad (1)$$

The discrete time version for implementation would be:

$$\theta_{k+1} = \theta_k + \omega_k T_s + w_k \quad \text{Discrete system model with noise} \quad (2)$$

where ω_k is angular velocity from 3-axis gyroscope, T_s is the sampling time of the process.

2.2 Measurement model

$$\theta_{acc,k} = \theta_k + v_k \quad \text{Measurement model} \quad (3)$$

θ_{acc} is the roll angle calculated from 3-axis accelerometer readings (note: we are treating the calculated angle as measurement, not the acceleration)

$$\theta_{acc} = \tan^{-1} \left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}} \right) \quad \text{Mode of measurement from the accelerometer} \quad (4)$$

2.3 Kalman Filter algorithm

2.3.1 Prediction

Propagate the system dynamics using the gyroscope measurements ($w_{gyro,k}$).

$$\hat{\theta}_{k+1|k} = \hat{\theta}_{k|k} + T_s w_{gyro,k} \quad \text{Prediction step} \quad (5)$$

Propagate the state variance ($\hat{\Sigma}$).

$$\hat{\Sigma}_{k+1|k} = \hat{\Sigma}_{k|k} + T_s^2 \sigma_w^2 \quad \text{A priori Covariance propagation} \quad (6)$$

σ_w^2 is the process noise variance.

2.3.2 Measurement update

Get the angle measurement from accelerometer.

$$\theta_{acc,k} = \text{atan2}(a_y, \sqrt{a_x^2 + a_z^2}) \quad \text{Measurement as stated earlier} \quad (7)$$

Update the angle estimate.

$$\hat{\theta}_{k+1|k+1} = \hat{\theta}_{k+1|k} + \frac{\hat{\Sigma}_{k+1|k}}{\hat{\Sigma}_{k+1|k} + T_s^2 \sigma_v^2} (\theta_{acc,k+1} - \hat{\theta}_{k+1|k}) \quad (8)$$

Gain of the filter A priori estimate Residual

σ_v^2 is the measurement noise variance. Some important terms in the above statement are:

- $\frac{\hat{\Sigma}_{k+1|k}}{\hat{\Sigma}_{k+1|k} + T_s^2 \sigma_v^2}$ is the Kalman gain.
- $\theta_{acc,k+1} - \hat{\theta}_{k+1|k}$ is called residual.

Update the state variance.

$$\hat{\Sigma}_{k+1|k+1} = \left(1 - \frac{\hat{\Sigma}_{k+1|k}}{\hat{\Sigma}_{k+1|k} + T_s^2 \sigma_v^2} \right) \hat{\Sigma}_{k+1|k} \quad \text{A posteriori covariance propagation} \quad (9)$$

3 Programming the ESP32

Steps with Resources:

1. Install Arduino IDE from this link.
2. To program ESP32 with Arduino IDE: Only read "Installing using Arduino IDE" of <https://docs.espressif.com/projects/arduino-esp32/en/latest/installing.html>. Follow the steps in this video (install latest versions only and choose "ESP32 Wrover Module" in board).

3. To interface LSM9DS1 and read sensor data, follow https://docs.arduino.cc/libraries/arduino_lsm9ds1/. Install `Arduino_LSM9DS1` library from library manager in Arduino IDE (go to Tools → Manage Libraries... OR simply open Library manager from the left pane).
4. You can use example codes for accelerometer and gyroscope interfacing. Go to File → Examples → `Arduino_LSM9DS1` → `SimpleGyroscope` (or `SimpleAccelerometer`).
5. `micros()` function can be used for measuring time.

If your COM port is not detected on connecting ESP32, you need to install USB to UART bridge driver (follow this video).

4 Assignment

- a. Implement roll angle estimation using gyroscope integration only. Use (2) (ignore noise term).
- b. Implement the estimation using accelerometer data only. Use (7).
- c. Implement the Kalman filtering algorithm as explained above. You can refer this video (different hardware setup is used in this tutorial).
- d. Observe the changes in Kalman filter estimate on changing σ_w , σ_v and initial Σ_k . Also observe effect of changing T_s .

Estimate of the roll angle from the three methods have to be displayed. Use serial plotter for visualization.